

Automated Temporal Localization of Behavior in Cichlid Fish with Few-Shot Trajectory Tokens

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Abstract. Understanding the brain requires studying its observable output: behavior. Such behavioral experiments in cichlid fish require manual annotations of aquarium video at real time playback speed. These annotations are time consuming to generate and isolated to single time points, which lack temporal context of the underlying behavior. We propose FSH-TAL, a temporal localization model that learns from sparse point annotations and predicts behavior-specific temporal segments on long-form video. On 17 hours of footage, FSH-TAL reaches state-of-the-art mAP (42.8% averaged across IoU 0.1-0.7).

Keywords: Behavior Classification · Temporal Action Localization

1 Introduction

Behavior is the observable output of the brain, whose development and function are shaped by genetics [19]. Cichlids, a family of fish, are a model system for studying social behaviors due to their distinct behavioral patterns [9] and social systems [14]. Furthermore, cichlids exhibit high rates of speciation, with more than 1,500 different species, providing an ideal system for comparing social behaviors amongst closely related species, with the potential to understand how genetics influences behavioral outputs [4, 6]. Quantifying distinct behaviors is therefore essential for understanding drivers of variation in behavioral patterns [16]. However, extracting this data from large volumes of video produced by behavioral experiments remains a major bottleneck [23].

In many animal behavioral experiments, experts manually annotate hours of recordings using tools such as BORIS [5], marking each behavior as a single point annotation. Automating the annotations of these recordings would save countless hours of human labor, accelerating ethological experiments [3]. Machine learning techniques are well suited for this task, as the two sources they need to imitate human predictions, data (videos) and labels (annotations), are readily available.

Current approaches use pose estimation techniques [15, 18] to train behavior classifiers [21, 22] to automate annotations. However, pose estimation requires high-quality video and human labels for anatomical segments, while standard filming techniques in aquatic environments introduce constraints such as frequent occlusion, low contrast, and small subjects. More generic action classification models [11, 12] can be paired with temporal action localization (TAL) models

[7, 13, 24, 25] to address these constraints and increase automation. These models use video transformers that extract features from raw video signal, sidestepping the aforementioned limitations of pose estimation. Still, most TAL models are tested on common databases [1, 8] that contain hundreds of hours of videos and hundreds of action classes. Behavioral studies may have as few as ten behaviors of interest and less than 20 hours of footage, introducing data limitations.

In this paper, we combine the fields of animal behavior and computer vision to create a tool that will accelerate discovery in behavioral experiments. Our contributions include: **(1)** a cichlid dataset spanning 6,517 point annotated behaviors across 17 $\sim$ 1 hour long recordings (Section 2); and **(2)** FSH-TAL, a model that can jointly localize and classify behaviors from this dataset with 42.8% average mAP, at least 15.5% better than previous methods with varied supervision. Our code and data will be released upon acceptance.

2 Dataset

We gathered data from behavioral experiments on the cichlid species *Astatotilapia burtoni*. We use 17 videos, each $\sim$ 1 hour long, containing two fish, and recorded at 25 FPS on the Raspberry Pi Camera Module 2 from a side profile of the tank. Expert observers annotated each video in BORIS [5] with time-stamped point events over a 6-class ethogram (Bite, Lead, Peck, Quiver, Chase/Charge, and Tilt), for a total of 6,517 event annotations. For all components of the model, we create 5 folds, training on 13-14 videos and testing on the remaining 3-4.

3 Method

Method Overview. Our model (Figure 1) has three key stages: **A)** First, we track points on the fish over time from a given input clip or video. **B)** Second, we train a Multiple Instance Learning (MIL) model using few-shot learning on the visual and point extraction features. **C)** Third, we use the frozen embeddings from the MIL classifier to train FSH-TAL, a temporal localization model.

Point Tracking. We first run Segment Anything Model 3 (SAM3) [2] with the text prompt “fish” and record the top two objects (i.e., male and female fish) ranked by temporal persistence. We extract 9 points from a 3×3 grid over each fish’s predicted mask, yielding 18 points per clip distributed over both fish in the video. Point trajectories are then generated by tracking points across frames.

MIL Classification. Our MIL model builds on Tokens [11], a classifier architecture that utilizes DINOv2 [17] features and tracked points fed into a video transformer. Our method improves on Tokens in two ways: 1) We generate trajectory points with SAM3 instead of the native CoTracker [10] to improve object segmentation in a noisy aquarium and 2) We adapt multi-label few-shot classification to address the limitations of our scarce dataset and courtship behaviors that co-occur (e.g., *Male Tilt* and *Female Peck*). Specifically, we train our models in 5-way 3-shot episodes using standard few-shot loss. Multi-labeled examples contribute to each of their target class’ prototypes.

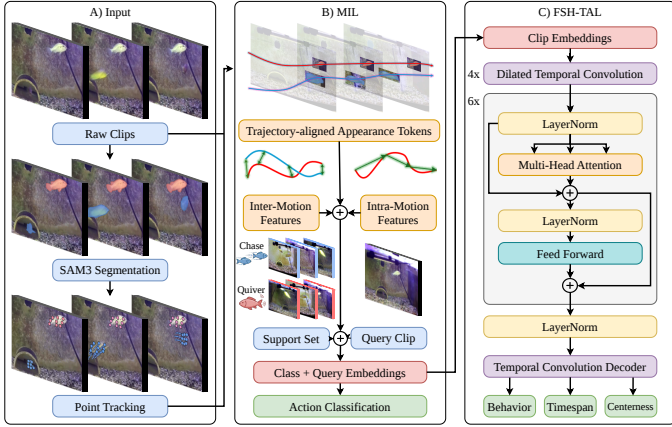


Fig. 1: FSH-TAL architecture. **A)** Points are generated from raw clips. **B)** We train a classification model using few-shot learning. Attention modules are run over all modules in **orange**. **C)** We train FSH-TAL using frozen embeddings from the MIL.

Temporal Localization. Deriving the full behavioral context from video footage requires temporal localization in addition to classification. FSH-TAL achieves both while still being trained on point annotations. Our model reinterprets each annotated point as a ± 2 s timespan, which serves as a weak temporal boundary label for observed behaviors. This time span is consistent with observed durations found across all videos. FSH-TAL then utilizes the frozen embeddings from the MIL, sampled at 4 FPS. Training uses a focal classification loss, Intersection over Union (IoU) regression loss, and centerness binary cross entropy loss on a candidate instance. Inference is done on rolling 90s windows with 45s overlap throughout the full recording. Predictions are decoded with a greedy temporal Non-Maximum Suppression (NMS) over each behavior.

4 Results

We evaluated all models on mAP@IoU averaged across all behaviors. The MIL model was evaluated on 4s rolling window clips with 2s overlap, after merging adjacent windows with the same behavior prediction (*MIL*). We then post-processed the MIL predictions with Outer-Inner Contrast scoring and temporal NMS (*MIL + post-proc*) [20] as a more thorough baseline. Table 1 shows that both models degraded at tighter thresholds, which is expected as MIL models cannot refine temporal boundaries of predictions beyond their input clip’s length.

We compare our methods with previously defined SOTA models for this task. *ASM-Loc* [7] is a weakly supervised TAL model which shows worse performance than our own *MIL + post-proc*. *HR-Pro* [25], which generates temporal segments from point annotations, and *ActionFormer* [24], a fully supervised method, both show better results at tighter IoU, but fall short of *MIL + post-proc*.

Table 1: Comparison of state-of-the-art methods with varying supervision on our dataset. [†]Not including parameters used to generate embeddings.

Method	#Params	mAP@IoU (%)				
		0.1	0.3	0.5	0.7	Avg
MIL classifier (baseline)	16.0M	25.7	12.0	4.6	0.5	10.5
MIL + post-proc	16.0M	39.1	20.7	3.1	0.1	14.9
ASM-Loc [7]	1.8M [†]	26.8	16.9	6.1	1.2	12.4
HR-Pro [25]	23.2M	33.6	21.7	7.8	1.2	15.6
ActionFormer [24]	27.3M [†]	35.1	32.2	25.4	14.0	27.3
FSH-TAL (Ours)	15.8M [†]	53.3	48.7	41.3	24.7	42.8

Table 2: Comparison of input features.

Table 3: Comparison of output heads.

		mAP@IoU (%)					mAP@IoU (%)		
MIL	SAM3	0.1	0.7	Avg	Centerness	Timespan	0.1	0.7	Avg
X	X	41.2	14.7	30.5	✓	X	52.2	24.8	42.2
✓	X	44.6	17.0	33.2	X	✓	54.3	24.5	43.1
✓	✓	53.3	24.7	42.8	✓	✓	53.3	24.7	42.8

at IoU=0.1. *ASM-Loc*, *ActionFormer*, and *FSH-TAL* all use classification-aware features which are extracted from the frozen MIL. Table 1 shows that *FSH-TAL* reaches **42.8% mAP** (average IoU 0.1–0.7), outperforming all tested methods.

Tables 2 and 3 show the impact of different architecture decisions within FSH-TAL. First, we compare the results of training FSH-TAL with raw DINOv2 [17] features, MIL features without points, and MIL features with SAM3 [2] points. The increased performance of SAM3 classification-aware embeddings (+9.6%) highlights the effectiveness of point tracking in the noisy environment. Additionally, we test FSH-TAL without the centerness and timespan heads, providing a fixed ± 2 s window for each prediction. Table 3 shows that removing the centerness head increased average mAP by 0.3%, but decreased mAP at high IoU. These two model heads may not be vital to our problem setting, but may still benefit animal behavior problems with variable timespans and ethograms.

5 Conclusion

We present a method for converting sparse point annotations into temporally localized behavior segments in long-form cichlid recordings. First, we contribute a cichlid behavior dataset spanning 6,517 point events across 17 recordings. Second, we introduce FSH-TAL (42.8% mAP), which outperforms all other tested TAL models. Our work provides a practical approach for converting routine point annotations into temporal behavior measurements, and enables streamlined data collection methods for broad genetic experiments.

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